Battery and EV Modelling for EDF

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This project develops a lithium-ion battery and electric vehicle model based on experimental cell data provided by EDF. An equivalent circuit approach is used to estimate state of charge and health, then scaled to a battery pack. The model is coupled with a WLTP-based vehicle dynamics simulation for a Renault Zoe, enabling estimation of energy consumption and driving range, with and without regenerative braking.

1 Introduction

The rapid growth of electric vehicles raises key challenges in battery sizing, energy efficiency, and ageing, making accurate vehicle and battery modelling essential. This project aims to estimate the driving range of a Renault Zoe under the WLTP cycle using a coupled vehicle and battery pack model.

To support this vehicle-level analysis, the work focuses on the electrical characterisation and modelling of a lithium-ion battery cell using experimental test data acquired at different ageing stages. Key battery indicators such as capacity, State of Charge (SOC), State of Health (SOH), Open Circuit Voltage (OCV), and internal resistance are extracted and used to identify the parameters of an equivalent electrical circuit model. This cell-level approach provides a robust foundation for battery pack modelling and electric vehicle simulations.

2 Cell Modelling

2.1 Data understanding and analysis

2.1.1 Description of the Experimental Dataset

The dataset we were provided with describes the electrical response of the studied lithium-ion battery at different ageing stages. Each test follows a structured current profile designed to understand and characterise the battery behaviour under dynamic loading [Figure 1]. The protocol begins with a charging phase to bring the battery to a defined state of charge, followed by a rest period at zero current to allow voltage relaxation. Then, a sequence of current pulses is applied to elicit a dynamic voltage response, resulting in progressive discharge. The test ends with a complete charge phase and a final full discharge of the battery.

2.1.2 Identification and Interpretation of Measured Quantities

To better interpret the experimental data, the following signals are visualised:

- voltage as a function of time,
- current as a function of time.

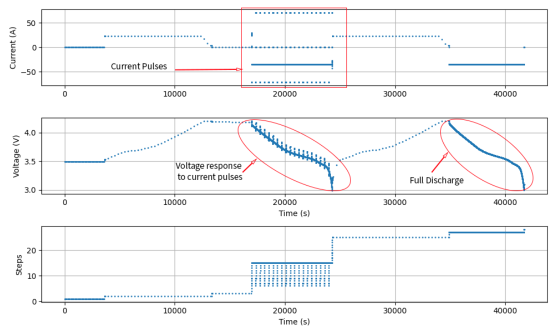

Figure 1: *Voltage and current as functions of time*

2.2 Electrical circuit modelling

2.2.1 Equivalent Electrical Circuit Model, Thevenin Model

To model the lithium-ion battery, we based our analysis on the Equivalent Circuit Model (ECM) described in RUOYU Xu's thesis *Lithium-ion battery modeling and SoC estimation*. We chose to represent the battery by a Thevenin electrical model [Figure 2]. We first worked on retrieving physical characteristics of our battery from the test data, mainly the State of Charge (SoC) and State of Health (SoH) during and through different tests.

A schematic representation of the equivalent circuit is presented, highlighting the different electrical components and their physical interpretation.

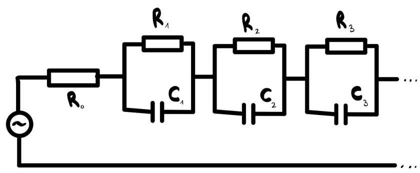

Figure 2: *Thevenin model (nth order)*

2.2.2 Associated Differential Equations

The differential equations governing the voltage dynamics of the model, providing the mathematical foundation for parameter identification and simulation. We worked on R0, and R1.

For R0:

$$\begin{cases} U_o = R_0 I_L \\ U_L = U_{OC} + U_o \end{cases}$$

For R1:

$$\begin{cases} I_L = \dfrac{U_1}{R_1} + C_1 \dfrac{dU_1}{dt} \\ U_L = U_{OC} - U_0 - U_1 \end{cases}$$

3 Cell Characterisation

3.1 Capacity

3.1.1 Physical Definition of Capacity

Battery capacity is defined as the total amount of electric charge that can be reversibly stored and delivered by the cell. It is commonly expressed in ampere-hours (Ah) and represents a key indicator of the battery's energy storage capability.

3.1.2 Capacity Estimation from Full Discharge

The battery capacity was estimated using the final phase of the experimental test, corresponding to a complete discharge of the cell. Assuming the battery to be fully charged at the beginning of this phase (SoC = 1), the measured current was integrated over time from full charge to full discharge.

The capacity was expressed in ampere-seconds (A·s) to ensure consistency with the time-based integration used for State of Charge computation. The numerical integration was performed using the trapezoidal rule implemented via NumPy's trapezoid function.

For the first ageing test, a capacity of 238 580.144 A·s, corresponding to 66.27 Ah, was obtained.

3.2 State of Charge (SOC)

3.2.1 Definition of SOC

The State of Charge (SOC) is a dimensionless indicator ranging from 0 to 1, which tracks the relative amount of charge remaining in the battery over time.

3.2.2 SOC Estimation from Experimental Data

The SOC was estimated using a Coulomb counting approach based on the measured current profile. The reference time t_0 was defined as the instant at which the battery is assumed to be fully charged (SOC = 1). The SOC at any time t was then computed according to:

$$\mathrm{SoC}(t) = 1 + \frac{1}{Q} \int_{t_0}^{t} I(\tau)\, \mathrm{d}\tau$$

where $I(t)$ is the measured current and Q is the previously estimated total capacity of the battery. This equation respects the sign convention for the current given by the cell data. The numerical integration was carried out using the trapezoidal rule implemented via NumPy's trapezoid function.

Figure 3 illustrates the evolution of the State of Charge computed using the trapezoidal Coulomb counting method. The SOC increases during charging phases and decreases linearly during discharge, reflecting the integration of the measured current over time.

The red dashed vertical line indicates the reference time t_0 at which the battery is assumed to be fully charged (SOC = 1). This instant is used as the initial condition for the Coulomb counting calculation. From this reference point, the SOC evolution is entirely driven by the measured current, leading to a progressive decrease during the subsequent discharge phases until full depletion.

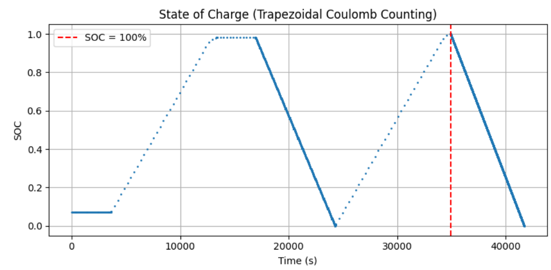

Figure 3: *State of Charge (Trapezoidal Coulomb Counting)*

3.3 State of Health (SOH)

3.3.1 Definition of SOH

The State of Health (SOH) is a dimensionless indicator that quantifies the degradation of a battery cell over time. It reflects the remaining usable capacity of the battery compared to its initial capacity in a fresh state. By definition, the SOH is equal to 1 (or 100%) for a new cell and decreases progressively as ageing mechanisms lead to irreversible capacity loss.

The SOH is defined as:

$$\mathrm{SOH} = \frac{Q_{\text{remaining}}}{Q_{\text{initial}}}$$

where $Q_{\text{remaining}}$ is the capacity measured at a given ageing stage and Q_{initial} is the reference capacity obtained for the fresh cell.

3.3.2 SOH Estimation Method

For each experimental test, the remaining capacity was estimated from the final phase of the protocol corresponding to a complete discharge of the cell. The SOH was then computed by normalising this capacity with respect to the initial capacity measured during test 0. This method provides a direct and physically meaningful measure of capacity fade induced by ageing.

3.3.3 SOH Evolution Across Ageing Tests

Figure 4 shows the evolution of the SOH for tests 0 to 4. Test 0 serves as the reference state, with an SOH equal to 1 by construction. A slight decrease in SOH is observed for test 1, indicating the onset of ageing and initial capacity loss.

As ageing progresses, tests 2 and 3 exhibit a more pronounced reduction in SOH, reflecting cumulative degradation mechanisms. The lowest SOH value is reached in test 4, highlighting a significant loss of usable capacity compared to the initial state. The monotonic decrease of SOH across the tests confirms the irreversible nature of battery ageing.

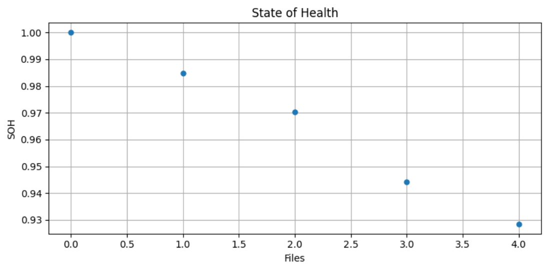

Figure 4: *State of Health evolution across ageing tests*

3.3.4 Impact of SOH on Cell Behaviour

The decrease in SOH directly affects the electrical behaviour of the battery cell. Capacity fade limits the amount of charge that can be delivered, while ageing is also expected to influence other parameters such as internal resistance and dynamic response. For this reason, SOH is used as a key input variable in subsequent analyses, particularly for resistance identification and equivalent circuit model parameterisation.

3.4 Open Circuit Voltage (OCV) analysis

The Open Circuit Voltage (OCV) is a fundamental characteristic of a lithium-ion battery, as it is strongly dependent on the State of Charge (SOC).

3.4.1 OCV Extraction from Rest Phases

To improve readability and focus on the most informative regions of the test, the OCV evolution was obtained by zooming into the part of the experiment corresponding to the current pulse sequence. This region is particularly relevant, as it spans a wide range of SOC values while containing multiple rest periods suitable for OCV extraction.

The OCV was extracted from voltage measurements during rest periods where the applied current is zero ($I = 0$). During these phases, the battery is assumed to be close to electrochemical equilibrium, allowing the measured terminal voltage to be considered a good approximation of the open-circuit voltage.

More precisely, for each zero-current plateau, the OCV value was taken as the voltage measured at the end of the rest period.

3.4.2 OCV–SOC Relationship

Once the OCV values were extracted, each value was associated with the corresponding SOC level computed via Coulomb counting. This enabled the construction of the OCV–SOC relationship, which captures the intrinsic dependence of the open-circuit voltage on the battery's state of charge.

The resulting curve exhibits the expected nonlinear behaviour, reflecting the underlying electrochemical processes occurring within the cell across different SOC regions.

3.4.3 Construction of the OCV(SOC) Function

To obtain a continuous representation of the OCV–SOC relationship, the discrete OCV values extracted from the rest plateaus were interpolated as a function of SOC. This interpolation step is necessary, as OCV values are only available at specific SOC points corresponding to the experimental rest periods.

Figure 5 presents the open-circuit voltage as a function of the State of Charge.

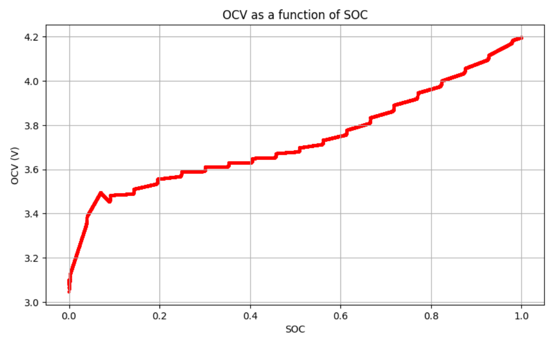

Figure 5: *OCV as a function of SOC*

The resulting curve shows a monotonic increase of OCV with SOC, with a steeper variation at low and high SOC levels and a flatter region in the mid-SOC range. This behaviour is characteristic of lithium-ion cells and reflects changes in the underlying electrochemical processes as the state of charge evolves. The interpolated OCV(SOC) function is subsequently used as an input to the equivalent circuit model to reproduce the voltage response under dynamic current conditions.

4 Internal Resistance Identification (R_0)

4.1 Identification Method for Internal Resistance

In order to model the electrical behaviour of the cell, it is necessary to identify the parameters governing its instantaneous and dynamic voltage response. This requires a detailed analysis of the experimental data obtained from repeated charge and discharge cycles. In particular, the voltage signal during discharge pulses exhibits a characteristic shape that can be exploited to extract physically meaningful parameters for the equivalent circuit model.

The identification of the internal ohmic resistance R_0 relies on the analysis of instantaneous voltage drops observed during current transients. These drops correspond to the purely resistive contribution of the cell and can be isolated from slower polarization effects through appropriate signal segmentation.

4.2 Analysis of R_0 Across Test Phases (A, B, C, D, E)

The identification process starts with a segmentation of the voltage signal during discharge events. By analysing the voltage response to current pulses, a recurring pattern can be observed, allowing the definition of characteristic time instants that correspond to distinct physical phenomena within the battery.

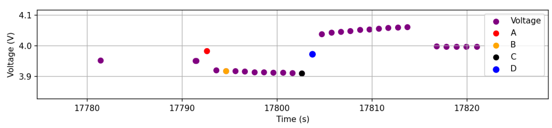

Figure 6: *Voltage response of the cell during a discharge pulse, highlighting the characteristic points A, B, C, and D.*

A: Instance just before the discharge pulse begins, detected as a local maximum of the voltage B: Instance marking the onset of the discharge pulse, identified as the breakpoint between two voltage slopes, corresponding to the instantaneous voltage drop C: Instance corresponding to the end of the relaxation phase, detected as a local minimum of the voltage D: Instance at which the applied current returns to zero ($I = 0$), marking the beginning of voltage relaxation back towards the open-circuit voltage

Identifying these segmentation points provides a clear framework for separating the different contributions to the voltage response and enables a consistent extraction of the internal resistance across the different regions of the test protocol (A, B, C, D, E).

4.3 Evolution of R_0 as a Function of SOC

Once the internal resistance has been extracted for each discharge pulse using the A–B voltage drop, each value of R_0 is associated with the corresponding State of Charge. This allows the analysis of the dependence of internal resistance on SOC, highlighting variations linked to the electrochemical state of the cell.

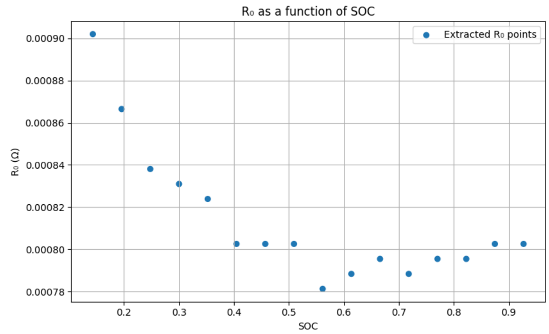

Figure 7: *Evolution of the internal ohmic resistance R_0 as a function of the State of Charge.*

The extracted R_0 values exhibit a moderate dependence on SOC, with higher resistance at low SOC and a lower, more stable behaviour in the mid-to-high SOC range.

To enable continuous modelling, the discrete R_0 points were interpolated to obtain an approximate $R_0(\mathrm{SOC})$ function, allowing the resistance to be evaluated at any SOC value between the measured points and used directly in the equivalent circuit model.

4.4 Model Validation

We validated the model by comparing its simulated voltage response with the measured terminal voltage during cell tests.

We found the simulated voltage through Kirchhoff laws and established the equation :

$$U_L = U_{OCV} + R_0 I_L$$

We see that the R is a scaling of the intensity, and follows the trend of the cell's response to a current impulse. The abrupt jump of voltage is well modelled by the order 0. However, the relaxation pattern of the voltage do not appear in this model, ad it does not include polarization.

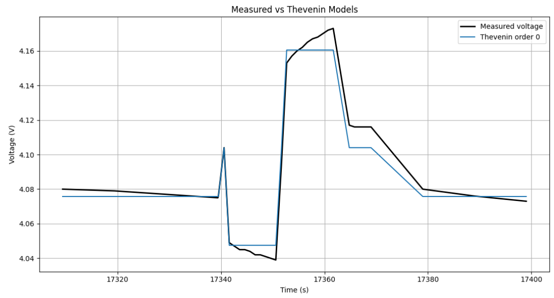

Figure 8: *Comparison between measured terminal voltage and simulated voltage using zero-order Thevenin models during a current pulse.*

5 First-Order Equivalent Circuit Model and Parameter Identification

In order to capture the dynamic behavior of the battery during current transients, a first-order Thevenin equivalent circuit model is considered. The RC branch accounts for polarization phenomena, which are responsible for the time-dependent voltage response observed during charge and discharge pulses.

During a discharge pulse, the terminal voltage exhibits a characteristic behavior that can be decomposed into several physical contributions. After the instantaneous ohmic voltage drop associated with R_0, the voltage continues to evolve more slowly due to polarization effects introduced by the RC branch. This regime corresponds to the time interval between points B and C, as identified in the signal segmentation procedure.

5.1 Exponential Relaxation Behavior

Under an imposed constant current I_L, the voltage contribution of the polarization branch satisfies the following differential equation:

$$C_1 \frac{dU_1(t)}{dt} + \frac{U_1(t)}{R_1} = I_L, \quad (1)$$

where $U_1(t)$ denotes the voltage across the RC branch. The solution of this first-order linear differential equation leads to an exponential response characterized by a time constant:

$$\tau_1 = R_1 C_1. \quad (2)$$

When the polarization voltage is observed during the relaxation phase, its temporal evolution follows an exponential decay. This behavior is clearly visible between points B and C of the voltage signal and provides a reliable means of identifying the time constant τ_1.

5.2 Exponential Fitting Procedure

To extract the time constant τ_1, an exponential fitting procedure is applied to the measured voltage during the relaxation phase. The voltage is first normalized according to:

$$y(t) = \frac{V(t) - V_C}{V_B - V_C}, \quad (3)$$

where V_B and V_C denote the voltages at points B and C, respectively. This normalization removes the influence of the voltage amplitude and isolates the exponential component of the response, yielding:

$$y(t) = e^{-t/\tau_1}. \quad (4)$$

This approach reduces the number of fitted parameters to a single scalar. Furthermore, the exponential fitting is applied to all detected relaxation segments within the cell data, allowing a more accurate estimation of τ_1.

5.3 Extraction of R_1 and C_1

Once the time constant τ_1 has been determined, the resistance R_1 can be obtained from the steady-state polarization voltage. Indeed, at point C, the polarization voltage reaches its asymptotic value:

$$U_1(\infty) = I_L R_1, \quad (5)$$

which leads to:

$$R_1 = \frac{V_B - V_C}{|I_L|}. \quad (6)$$

Finally, the capacitance C_1 is computed directly from the definition of the time constant:

$$C_1 = \frac{\tau_1}{R_1}. \quad (7)$$

This identification procedure allows the parameters of the first-order Thevenin model to be extracted in a physically meaningful manner. Each parameter is identified in the regime where its influence on the voltage response is dominant, ensuring consistency between the measured data and the equivalent circuit representation.

5.4 Model Validation

The identified first-order Thevenin model is validated by comparing its simulated voltage response with the measured terminal voltage during current transients. For reference, a zero-order Thevenin model (pure ohmic behavior) is also considered.

6 EV Modelling

2.1 Theory and Reference framework

2.1.1 Reference Vehicle: Renault Zoe (ZE 40)

For this study, we selected the Renault Zoe R110 ZE 40. This choice is motivated by the abundance of resources on this model as well as by the project objective to size a battery for a standard urban vehicle.

Here are a few of the vehicle's technical characteristics, recovered from the manufacturer:

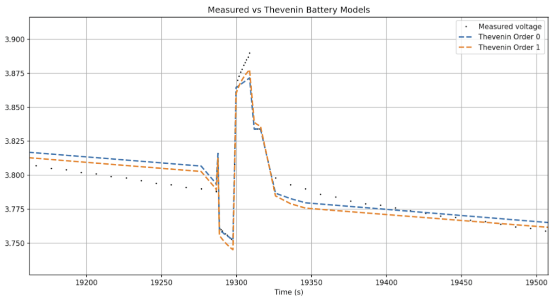

Figure 9: *Comparison between measured terminal voltage and simulated voltage using zero-order and first-order Thevenin models during a current pulse. The first-order model captures the relaxation dynamics more accurately due to the inclusion of polarization effects.*

Motor: The R110 is a synchronous electric motor delivering a maximum power of 80kW and a torque of 225Nm.

Battery: The ZE 40 battery pack has a usable capacity of 41kWh. It is composed of 192 LG Chem NMC (Nickel-Manganese-Cobalt) cells, arranged in 96 groups in series and 2 in parallel (96s2p), delivering a nominal voltage of approximately 350-360 volt.

Target Performance: This configuration allows for a theoretical range of 300km under the WLTP cycle, which serves as the validation target for our model.

2.1.2 The Standardized Cycle: WLTP

To estimate the energy consumption and range of the vehicle, we use the WLTP (Worldwide Harmonized Light Vehicles Test Procedure). This standard, mandated in Europe since 2017, defines a strictly controlled test procedure to ensure reproducible results.

The core of the simulation relies on the WLTC (Worldwide Harmonized Light-duty Test Cycle) speed profile. It consists of a 30-minute (= 1,800 seconds) driving cycle covering 23.25km, divided into four distinct phases representing different driving environments, ranging from urban stop-and-go traffic (Phase 1) to high-speed motorway cruising (Phase 4).

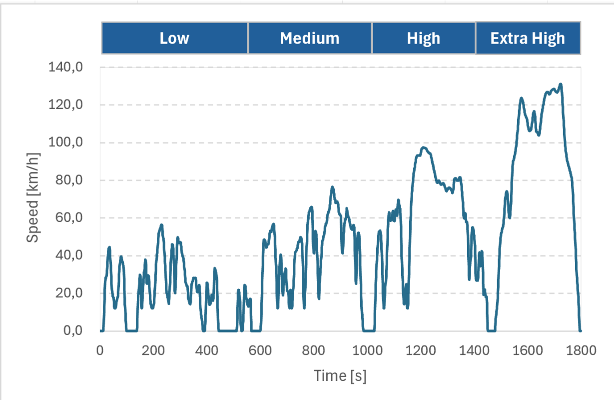

Figure 10: *Velocity profile of the WLTC Class 3b cycle*

We specifically use the Class 3b cycle, applicable to vehicles with a high power-to-mass ratio and a maximum speed above 120km/h, which corresponds to the Renault Zoe R110. This speed profile (v vs t) provides the input data to calculate the instantaneous mechanical power required at the wheels.

2.2 Modelling the EV: inventory of forces & assumptions

2.2.1 Inventory of the forces acting on the car

According to Newton's second law, the motion of an object is governed by the sum of the external forces applied to it. During a WLTP test, a Renault Zoe is subjected to several forces that oppose the vehicle's motion, while others contribute to it.

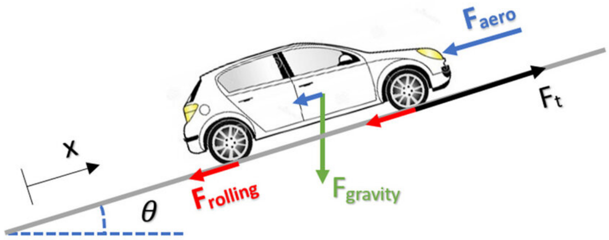

Figure 11: *Forces Acting on a Vehicle*

The forces considered are the following:

Aerodynamic drag force

$$F_d = -\frac{1}{2}\,\rho\,A_f\,C_d\,v^2$$

Rolling resistance force

$$F_r = -C_{rr}\,m\,g\cos(\theta)$$

Gravitational force

$$F_g = -m\,g\sin(\theta)$$

Traction force

$$F_t$$

where ρ is the air density, A_f the frontal area of the vehicle, C_d the aerodynamic drag coefficient, C_{rr} the rolling resistance coefficient of the tyres, and θ the road slope angle.

Applying Newton's second law yields:

$$\sum F = ma = F_d + F_r + F_g + F_t$$

where a is the acceleration of the vehicle.

2.2.2 Assumptions of the model

To model the Renault Zoe R110 ZE 40, we rely on publicly available technical data and standard WLTP assumptions. According to WLTP regulations, the vehicle mass corresponds to the curb mass plus a standard driver mass of 75 kg.

The parameters retained for the model are:

- Vehicle mass: $m = 1577$ kg
- Rolling resistance coefficient (Class B tyres): $C_{rr} = 0.008$ (typical range: 0.007–0.009)
- Aerodynamic parameter: $A_f C_d = 0.75$

During the WLTP test, the road is assumed to be flat. Consequently, the slope angle is set to $\theta = 0$, which implies $F_g = 0$.

Under these conditions, Newton's second law simplifies to:

$$\sum F = ma = F_d + F_r + F_t$$

To compute the mechanical power at the wheels and the resulting energy consumption, additional assumptions are required regarding the drivetrain efficiency and auxiliary power demand, auxiliaries being all functions of the car that don't make it move forward. The following values are adopted:

- Drivetrain efficiency: $\eta = 0.90$
- Auxiliary power demand: $P_{\text{aux}} = 500$ W

2.3 Model and Results

2.3.1 How the EV model works & results

Using the inventory of forces and the assumptions explained previously, we then proceeded to compute the mechanical and electrical power required for the car to maintain the various speeds of the WLTP test. From the inventory of forces developed earlier, we isolate the traction force:

$$F_t = -F_d - F_r + F_i$$

From this traction force, we determine the mechanical power required at the wheels to make the car move:

$$P_{\text{mech}} = F_t \, v$$

Once the mechanical power is known, the electrical power required from the motor is computed using the drivetrain efficiency η. This electrical power corresponds to the power that must be delivered by the battery for the wheels to receive the necessary mechanical power:

$$P_{\text{elec}} = \frac{P_{\text{mech}}}{\eta} + P_{\text{aux}}$$

Using the velocity and acceleration profiles of a standard WLTP Class 3 test, we can plot an approximation of the electrical power required by the Renault Zoe over time.

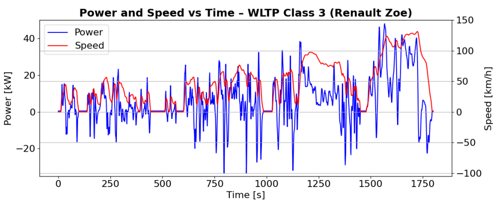

Figure 12: *Power and Speed over time - WLTP test*

Our model is coherent with the expected physical behaviour. The power becomes negative during deceleration phases and positive during acceleration phases, and it remains equal to the auxiliary power demand when the vehicle speed is zero. The obtained power values also remain within the expected range based on available technical documentation.

2.3.2 Verification and comments

To verify our model, we calculated an estimation of the autonomy of the Zoe and compared it to the official WLTP range of 300 km. The instantaneous electrical power demand over the WLTP cycle is integrated over time to compute the total energy consumption:

$$E_{\text{WLTP}} = \sum_k P_{\text{batt}}(t_k)\,\Delta t$$

The travelled distance over the WLTP cycle is obtained by integrating the vehicle speed over time:

$$d_{\text{WLTP}} = \sum_k v(t_k)\,\Delta t$$

The average energy consumption of the vehicle is then calculated as:

$$C = \frac{E_{\text{WLTP}}}{d_{\text{WLTP}}}$$

Assuming a usable battery energy of $E_{\text{batt}} = 41$ kWh for the Renault Zoe R110 ZE 40, the vehicle range is then estimated as:

$$\text{Range} = \frac{E_{\text{batt}}}{C}$$

Our model yields an estimated range of 246.8 km, corresponding to an accuracy of 82.2% with respect to the official WLTP value.

In addition to the base model, a version including regenerative braking was implemented. When the mechanical power at the wheels is positive, the electrical power demand is computed as in the previous case. When the mechanical power becomes negative, a fraction of this power is assumed to be recovered and fed back to the battery through regenerative braking, with an efficiency factor $\eta_{\text{regen}} = 65\%$. The resulting battery power is thus written as:

$$P_{\text{elec, regen}} = \begin{cases} \dfrac{P_{\text{mech}}}{\eta} + P_{\text{aux}}, & P_{\text{mech}} \geq 0 \\ \eta_{\text{regen}}\, \eta \, P_{\text{mech}} + P_{\text{aux}}, & P_{\text{mech}} < 0 \end{cases}$$

It should be noted that this approach is optimistic, as regenerative braking only recovers power during actual braking events. In this model, however, all deceleration phases are treated as braking phases, due to the lack of publicly available data distinguishing the two. As a result, the recovered energy is likely overestimated.

With this regenerative braking model, the estimated range increases to 287.2 km, corresponding to an accuracy of 95.7% with respect to the official WLTP value. This result highlights the significant impact of regenerative braking on vehicle autonomy, while also underlining the limitations of the present modelling assumptions.

It should be noted that this verification part of the model does not take into account the SoC degradation of the battery. The model therefore remains macroscopic-level.

3 Battery Pack Modelling and Coupling with the EV Model

3.1 Battery pack construction

The experimental characterisation was first carried out at the *cell level*, where three key quantities were identified as functions of the state of charge (SoC): the open-circuit voltage $\mathrm{OCV}(\mathrm{SoC})$, the internal resistance $R_0(\mathrm{SoC})$, the usable capacity Q_{cell}.

To model the full battery, we created a pack by assembling cells as follows:

- $m = 111$ cells in series,
- $n = 2$ cells in parallel.

The number of cells in series n is determined from the pack voltage requirement using

$$n = \frac{V_{\text{pack}}}{V_{\text{cell}}} = \frac{400}{3.6} = 111,$$

which ensures that the nominal pack voltage matches the target vehicle voltage that is of 400V for the Renault Zoe. The number of cells in parallel m is obtained from the energy requirement of the battery pack,

$$E_{\text{pack}} = n\, m\, E_{\text{cell}},$$

with $E_{\text{cell}} = 222$ Wh and $E_{\text{pack}} = 41$ kWh. Which links the total pack energy to the energy stored in each individual cell. Rounding to the nearest integers yields a physically realizable battery pack configuration, here $n = 111$ cells in series and $m = 2$ cells in parallel, providing both the required voltage and energy.

Using equivalent circuit electrical laws, the equivalent pack parameters were derived as follows.

The pack open-circuit voltage is given by

$$\mathrm{OCV}_{\text{pack}}(\mathrm{SoC}) = m\, \mathrm{OCV}_{\text{cell}}(\mathrm{SoC}).$$

The pack internal resistance is

$$R_{\text{pack}}(\mathrm{SoC}) = \frac{m}{n}\, R_{0,\text{cell}}(\mathrm{SoC}).$$

Finally, the pack capacity is

$$Q_{\text{pack}} = n\, Q_{\text{cell}}.$$

3.2 EV power demand and battery coupling

The electric vehicle (EV) model provides the electrical power demand $P_{\text{elec}}(t)$ over a WLTP driving cycle, derived from longitudinal vehicle dynamics (aerodynamic drag, rolling resistance and inertia) and drivetrain efficiency.

The following sign convention was adopted, as it follows the conventions of the tests' data:

- $P_{\text{elec}} > 0$: the vehicle requires traction power,
- $I > 0$: the battery is charging,
- $I < 0$: the battery is discharging.

Since the battery supplies power during traction phases, the battery power is defined as

$$P_{\text{batt}}(t) = -P_{\text{elec}}(t).$$

3.3 Order-0 electrical battery model

At order 0, the battery is modelled as an ideal voltage source $\mathrm{OCV}_{\text{pack}}$ in series with an internal resistance R_{pack}. The terminal voltage is therefore

$$V = \mathrm{OCV}_{\text{pack}} + R_{\text{pack}} I,$$

due to the convention on I. The electrical power balance at the battery terminals gives:

$$P_{\text{batt}} = VI = (\mathrm{OCV}_{\text{pack}} + R_{\text{pack}} I)I.$$

This leads to the quadratic equation:

$$R_{\text{pack}} I^2 + \mathrm{OCV}_{\text{pack}} I - P_{\text{batt}} = 0.$$

Solving for the current yields:

$$I = \frac{-(\mathrm{OCV}_{\text{pack}} + \sqrt{\mathrm{OCV}_{\text{pack}}^2 + 4R_{\text{pack}} P_{\text{batt}}})}{2R_{\text{pack}}},$$

where the negative root is selected to ensure a physically consistent current sign during discharge.

3.4 SoC update

The state of charge is updated using Coulomb counting:

$$\mathrm{SoC}_{k+1} = \mathrm{SoC}_k + \frac{I_k\, \Delta t}{Q_{\text{pack}}},$$

where Δt is the WLTP time step. The SoC is constrained to the interval $[0, 1]$ to enforce physical bounds.

3.5 WLTP cycles and vehicle autonomy

The WLTP power profile is applied repeatedly until the battery reaches a minimum state of charge. Each WLTP cycle corresponds to a distance of approximately 23.25 km. The total driving range is therefore estimated as

$$\text{Autonomy} = N_{\text{cycles}} \times 23.25 \text{ km},$$

where N_{cycles} is the number of completed WLTP cycles before battery depletion.

4 Results and Limitations

4.1 Results

Manufacturer-reported vehicle performance entailed an autonomy of 300km for the Renault Zoe. With our model, a order 0 and at perfect state of health, we achieved an autonomy of 300.9km for approximately 13 cycles of WLTP, that is 99% of accuracy for the merged model.

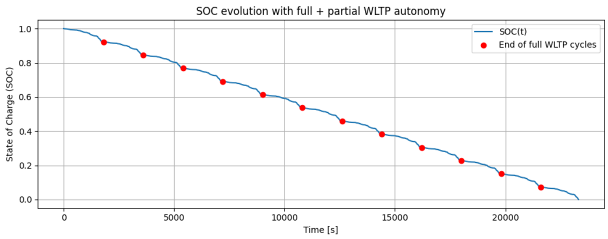

Figure 13: *Plot of the SoC response from full charge to full discharge of the battery at ageing stage 1 to calculate autonomy, model based on order 0 approximations*

Despite the good agreement obtained with manufacturer-reported autonomy, several limitations and simplifying assumptions of the model must be highlighted.

First, the treatment of regenerative braking remains simplified. All phases of decreasing vehicle speed were interpreted as braking events, and a constant regeneration efficiency was applied. In reality, regenerative braking depends on braking demand, motor limitations, battery acceptance power, and control strategies, making this assumption optimistic and likely to overestimate recovered energy.

Second, the WLTP autonomy was computed by terminating the simulation when the state of charge reached a threshold of $\mathrm{SOC} = 0.01$. While this avoids numerical instability, it does not fully reflect real vehicle behaviour, where a safety margin typically prevent operation near complete discharge.

Third, the estimation of the first-order dynamic parameters R_1 and C_1 relied on relaxation curves extracted around current steps and interpolated across operating points. This approach assumes a single dominant time constant and may introduce accumulated approximation errors when scaling from cell-level dynamics to pack-level behaviour.

Finally, the dependence of electrical parameters on state of charge was handled through interpolation of experimentally extracted points. While this ensures continuity of the model, it smooths local discontinuities, potentially masking sharp variations in battery behaviour.

Overall, these assumptions were necessary to maintain tractability and coherence of the coupled vehicle–battery model, but they define the validity domain of the results and point toward possible improvements for higher-fidelity modelling.

4.2 Ageing of the battery

Battery ageing was incorporated through the evolution of experimentally identified cell parameters across different ageing stages. A reduction in usable capacity and an increase in internal resistance were observed as the state of health decreased, directly impacting vehicle autonomy. For a given WLTP power demand, ageing leads to higher current levels, increased resistive losses, and a faster discharge, thereby reducing the number of autonomy. These results highlight the strong coupling between cell-level degradation mechanisms and vehicle-level performance, and confirm that battery ageing is a key determinant of electric vehicle range.

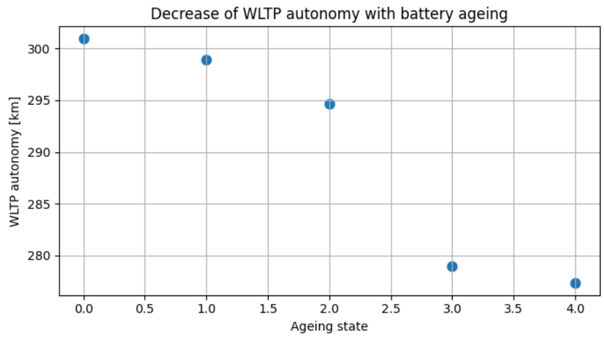

Figure 14: *Plot of the autonomy of the car decreases through ageing of the battery*

4.3 Sensitivity Analysis

To assess the robustness of our results and identify the primary drivers of vehicle autonomy, a sensitivity analysis was performed on key model parameters. This analysis evaluates how variations in vehicle mass, aerodynamic and rolling coefficients, auxiliary loads, and powertrain efficiencies influence the estimated driving range.

4.3.1 Impact of Vehicle Physics ($C_{rr}, C_d A_f, m$)

The resistive forces opposing motion – rolling resistance and aerodynamic drag – scale differently with speed.

- **Rolling Resistance (C_{rr}):** The rolling resistance force is defined in our model as $F_r = C_{rr} m g \cos(\theta)$. Since the WLTP cycle contains significant periods of low-to-medium speed driving (Phases Low and Medium), the model is moderately sensitive to the tire friction coefficient. A deterioration in tire pressure or road quality increasing C_{rr} from the nominal 0.008 would uniformly increase consumption across all speeds.
- **Aerodynamics ($C_d A_f$):** The drag force increases quadratically with velocity ($F_d \propto v^2$). While less critical in urban phases, this parameter becomes the dominant factor during the "Extra High" phase of the WLTP cycle (speeds up to 130 km/h). Small

improvements in the drag coefficient yield disproportionately high gains in highway autonomy compared to urban range.

- **Vehicle Mass (m):** Mass directly impacts both rolling resistance and the inertial force $F_{inertia} = ma$ required to accelerate. Given the frequent stop-and-go nature of the WLTP cycle, the inertia term is significant. However, increasing the battery size to improve range incurs a "mass penalty": the added weight increases consumption, creating a diminishing return on the added capacity.

4.3.2 Powertrain and Auxiliary Efficiencies

- **Traction Efficiency (η):** The model assumes a constant drivetrain efficiency of $\eta = 0.90$. This parameter has the **highest sensitivity** regarding energy consumption. Since it acts as a divisor in the power equation $P_{elec} = P_{mech}/\eta$, a 1% decrease in efficiency leads to a direct and slightly greater than 1% increase in battery energy depletion.
- **Regenerative Efficiency (η_{regen}):** Our results show that regenerative braking extends the range from 246.8 km to 287.2 km, a gain of approximately 16%. Consequently, the sensitivity to η_{regen} is secondary compared to traction efficiency. A slight variation in the assumed 65% regeneration factor would only scale the recovered energy portion, resulting in a minor impact on total range compared to traction efficiency changes.
- **Auxiliary Power (P_{aux}):** The constant auxiliary load is set to 500 W. Over the 30-minute WLTP cycle, this consumes roughly 250 Wh. While this represents a small fraction of the total 41 kWh capacity, its relative impact increases significantly in low-speed, high-traffic scenarios (urban driving) where the distance covered per unit of time is low, making the fixed P_{aux} a larger percentage of the total Wh/km consumption.

4.3.3 Battery State of Health (SOH)

The State of Health is the most critical variable defining the useful life of the vehicle.

- **Capacity Fade:** As defined previously, SOH linearly scales the usable capacity $Q_{remaining}$. A drop in SOH from 100% to 80% results in an immediate 20% loss in theoretical range.
- **Internal Resistance Increase:** As observed in the cell characterisation, ageing correlates with an increase in internal resistance R_0. This creates a secondary, non-linear sensitivity: as R_0 rises, resistive heat losses ($P_{loss} = R_{pack}I^2$) increase. This further reduces the effective energy delivered to the motor, causing the "practical" range to decrease faster than the loss of capacity alone would suggest.

Table 1: *Summary of Sensitivity Analysis*

Parameter	Sym.	Sens.	Dominant Regime
State of Health	SOH	**Very High**	All Phases
Traction Efficiency	η	**High**	All Phases
Aerodynamics	C_dA_f	**High**	Highway (Extra High)
Vehicle Mass	m	**Medium**	Acceleration (Urban)
Regen Efficiency	η_{reg}	**Medium**	Deceleration phases
Auxiliary Power	P_{aux}	**Low**	Traffic / Low speed

4.4 User Interface and Interactive Experimentation

4.4.1 Objective

The objective of the user interface is to bridge the gap between the raw computational logic and practical usability. While the code handles the physical and algorithmic complexity, the interface allows users to interact with the model without manipulating the underlying codebase. This decoupling is essential for making the project a dynamic exploratory tool.

4.4.2 Method and Key Functionalities

We selected Streamlit as our framework for its capability to rapidly turn Python scripts into shareable web applications. Unlike traditional web frameworks, Streamlit integrates directly with our data science libraries (Pandas, NumPy, etc.). This minimizes the context switching between the core logic and the UI, ensuring that the visual tool remains strictly synchronized with the analytical backend.

The interface is designed around an iterative experimentation loop, focusing on the following core capabilities:

- **Granular parameter control:** The sidebar controls allow for precise tuning of input variables (e.g., rolling resistance coefficient, efficiency). This enables sensitivity analysis, allowing the user to immediately observe how minor fluctuations in input parameters impact the global result.
- **Real-time visualization & feedback:** The central panel renders results dynamically. We leverage Matplotlib to generate interactive visualizations. The application uses reactive programming principles: as soon as a user updates a parameter, the relevant codes are triggered, and the visual output is refreshed.

5 Conclusion

This study presented a coupled modelling framework for lithium-ion battery behaviour and electric vehicle energy consumption under standardized driving conditions. An equivalent circuit model was used to estimate the state of charge and voltage response from experimental cell data at different ageing stages of the battery, while a force-based vehicle model simulated the Renault Zoe over the WLTP cycle. The resulting autonomy estimates were in good agreement with official values, particularly when regenerative braking was taken into account.

The end goal of this work was the coupling of the vehicle energy demand with a physically consistent battery pack model. Cell-level electrical characteristics were scaled to pack level through series and parallel configurations, allowing the pack voltage, internal resistance, and available capacity to be directly derived from experimental measurements. The battery current was obtained by solving the power balance imposed by the vehicle demand, ensuring consistency with fundamental electrical and mechanical laws.

The final integrated model enabled continuous tracking of the state of charge over complete and partial WLTP cycles, leading to realistic autonomy predictions. The use of a clear current sign convention and an explicit treatment of regenerative braking further improved the robustness and predictive capability of the model.

6 Credits

6.1 Authors of the template

Vel (`vel@LaTeXTemplates.com`), Frits Wenneker, authors of the Wenneker article template (downloaded from `http://www.LaTeXTemplates.com`).

6.2 License

6.3 Bibliography

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